

theme \ year	2023	2024	2025	2026 H1	cutoff
Foundations & native training	• BitNet - BitLinear, 1-bit from scratch • TSLD - first ternary QAT of a large generative LM	• BitNet b1.58 - ternary, FP16 parity from ~3B • Spectra / TriLM - open ternary suite 99M-3.9B • MatMul-free LM - removes the attention matmul	• BitNet b1.58 2B4T - first open native 2B model, 4T tok • BitNet a4.8 - 4-bit activations • BitNet v2 - Hadamard rotation, native A4	• Sparse-BitNet - N:M sparsity + ternary • Sherry - 1.25-bit packing • NanoQuant / LittleBit - sub-1-bit	Literature cutoff 30 June 2026 Later work tracked in the online living appendix
Post-training routes toward one bit	• PB-LLM - first LLM binarization for compression	• BiLLM (1.08-bit) • BinaryMoS • DB-LLM • OneBit • ARB-LLM • FBI-LLM	• PTQTP • PT2-LLM • Tequila • PTQ1.61 • ICQuant	• HARP • TWLA • BWLA - first W1A6 • influence-weighted Walsh rotations	
Systems & inference kernels	–	• bitnet.cpp / 1-bit AI Infra 1.1 - CPU ternary kernels	• Spectra 1.1 TriRun - GPU ternary kernel	–	
Hardware accelerators	–	–	• PIM-LLM • BitROM • TeLLMe • TerEffic	• TOM - compute-in-ROM • Platinum - LUT-ASIC • T-SAR - SIMD-ISA • Vec-LUT	
Applications & extensions	–	–	• BitVLA (robotics) • BitTTS • one-bit ASR • BitNet Distillation • MoTE (ternary MoE)	• Ternary Mamba • BitNet Text Embeddings • Fairy2i (real -> complex 2-bit)	
Theory & scaling analysis	–	• Scaling Laws for Precision • QiD (undertrained-LLM favoring) • theory of 1-bit NNs	• QuEST (stable W1A1) • CAGE (curvature-aware STE) • ParetoQ (unified 1-4 bit)	• “Fitting is not enough” (smoothness) • extreme-low-bit reasoning failure modes	